Lab: Derivation Exercise — Predictive Coding Message Passing

Objective

Derive the predictive coding equations step by step for increasingly complex generative models, connecting the mathematics to the neural architecture.

Part 1: Single-Level Gaussian Inference

Goal: Derive belief updating for the simplest possible model.

Generative model: $p(s) = N(s; \mu_0, \sigma_0^2)$, $p(o|s) = N(o; s, \sigma\_o^2)$

Recognition density: $q(s) = N(s; \mu, \sigma^2)$

Task:

1. Write the free energy $F = -E\_q[\ln p(o|s)] + D\_KL[q(s) \| p(s)]$
2. Expand each term analytically using Gaussian integrals
3. Compute $\partial F/\partial\mu$ and set to zero → derive μ^*
4. Show that $\mu^* = (\sigma\_o^2 \cdot \mu_0 + \sigma_0^2 \cdot o) / (\sigma\_o^2 + \sigma_0^2)$ — a precision-weighted average of prior and observation

Write your response here

Part 2: Two-Level Hierarchical Model

Goal: Derive message passing in a hierarchy.

Generative model:

- Level 2 prior: $p(s_2) = N(s_2; \mu_{20}, \sigma_{20}^2)$
- Level 2 → Level 1: $p(s_1|s_2) = N(s_1; g_2(s_2), \sigma_2^2)$ where $g_2(s_2) = A \cdot s_2$ (linear)
- Level 1 → Observations: $p(o|s_1) = N(o; g_1(s_1), \sigma_1^2)$ where $g_1(s_1) = C \cdot s_1$ (linear)

Recognition density: $q(s_1) = N(\mu_1, \Sigma_1)$, $q(s_2) = N(\mu_2, \Sigma_2)$

Task:

1. Write the free energy as a sum of terms across levels
2. Compute $\partial F/\partial \mu_1$ and $\partial F/\partial \mu_2$
3. Identify the prediction errors $\varepsilon_1 = o - C \cdot \mu_1$ and $\varepsilon_2 = \mu_1 - A \cdot \mu_2$
4. Show that each level's update depends on bottom-up prediction errors (from below) and top-down predictions (from above)

Write your response here

Part 3: Nonlinear Extensions

Goal: Extend to nonlinear generative mappings.

Replace $g_1(s_1) = C \cdot s_1$ with a nonlinear function $g_1(s_1)$ (e.g., sigmoid, polynomial).

Task:

1. Derive the modified prediction error equation using the Jacobian $\partial g_1/\partial s_1$
2. Show that the gradient $\partial F/\partial \mu_1$ now involves the Jacobian transposed times the precision-weighted prediction error
3. Explain why this requires the Laplace approximation (evaluating the Jacobian at the current estimate μ_1)

Write your response here

Part 4: Generalized Coordinates

Goal: Extend predictive coding to temporal dynamics using generalized coordinates.

In generalized coordinates, the state vector is extended to include the state and its temporal derivatives: $\tilde{x} = [x, x', x'', ...]$

Task:

1. Write the generalized state-space model: $p(o|\tilde{s})$ and $p(\tilde{s})$ in generalized coordinates
2. Show that the free energy gradient in generalized coordinates has the form: $d\tilde{\mu}/dt = D\tilde{\mu} - \partial F/\partial\tilde{\mu}$, where D is the shift operator
3. Explain how this enables the system to predict the temporal evolution of sensory input

Write your response here

Part 5: Synthesis

In 200 words, explain how Parts 1-4 build a complete mathematical framework: from single-level inference to hierarchical message passing to temporal prediction — and how this framework maps onto the cortical hierarchy.

Write your response here

Lab Summary

Part	Skill Developed	Mathematical Result
1	Gaussian inference	Precision-weighted averaging (Bayesian updating)
2	Hierarchical derivation	Bottom-up/top-down message passing
3	Nonlinear extension	Jacobian-based prediction error propagation
4	Temporal dynamics	Generalized coordinates and prediction
5	Integration	Connecting math to neural predictive coding